09 SL First-Pass Results (local-data-only)

Honest first-pass triangulation across 3 local data layers for the Paper 9 candidate panel (36 genes, 8 target classes). No DepMap/ PRISM/GDSC/CTRP download. No GPU. No wet-lab. No Paper 1/2/3/4 manuscript touch. No voice-protected prose.

Status: first-pass complete

Hypothesis-generating only

DepMap deferred (gating analysis)

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Anchor: **paper9_plan.md**

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I. 한 줄 결론

3 local data layer triangulation 으로 Paper 9 후보 우선순위가 update 됐다. KCNN4 (DGE q=4e-32, log2FC=-4.00, CCLE r=+0.29) 가 lead candidate. **OSMR + IL6R** (cytokine receptor blockade) 이 JAK/STAT axis 의 진짜 SL window — STAT3 자체가 아니다. **SFK (LYN/SRC/FYN)** coherent class signal. **ATR Cox HR=6.88 p=0.024 + CHEK2 HR=0.48 p=0.040** 이 state-conditional DDR vulnerability fingerprint. **TYK2** 는 target 아니라 protective modifier 로 reframe. **TACSTD2 (TROP2)** 는 carved-out non-SL. SL claim 은 DepMap 검증 후에만.

LEAD CANDIDATE	TCGA COX EVENTS	DGE COVERAGE	CCLE COVERAGE
KCNN4	~18	36/36	8/36
DGE q=4e-32, log2FC=-4.00 (DM1 ↑); CCLE r=+0.29	n=560, event rate 3.2% (underpowered)	all candidates measured (51,711-gene panel)	local thyroid 4007-gene panel n=13

PROBLEM

Paper 9 plan 은 모든 candidate / 모든 class 를 동등하게 다룸. 실제 데이터 layer 에서 어디가 강하고 약한지 first-pass 로 본다.

IDEA

local data 만으로 3 layer (CCLE n=13 / DGE 51,711 / TCGA Cox n=560) 를 같이 본다. DepMap 다운로드 없이 directional ranking 가능.

DEFENSE

DepMap 검증 전이므로 SL pair claim 안 함. Cox underpowered 이므로 directional prior 로만. CCLE n=13 은 sanity check.

DECISION

Paper 9 plan 의 § 6 candidate target classes update — KCNN4/OSMR/IL6R/SFK/ATR-CHEK2 promote, pure STAT3 / TYK2 demote/reframe.

2. 비전공자 배경

본 페이지는 Paper 9 (Ruppin-style SL vulnerability map) 의 **first-pass** 분석 결과다. "first-pass" 는 모든 분석을 끝낸 게 아니라 – 가능한 local data 만 써서 우선순위가 어디로 가는지 본 것. DepMap CRISPR / PRISM IC50 / wet-lab 검증은 모두 deferred.

3 개 layer 를 같이 본다. **(1) CCLE n=13** 갑상선 cell line 발현 panel – 작아서 통계 검정 안 되지만 axis 가 진짜 lineage-collapsed state 인지 확인하는 sanity floor. **(2) DM1 vs DM2 DGE** TCGA-THCA bulk 51,711 유전자 – 36 candidate 모두 측정됨, q-value 가장 분명한 layer. **(3) TCGA-THCA Cox** univariate HR – n=560 으로 cohort 는 크지만 OS event 가 18 개로 underpowered, "p<0.05 안 나온 것 = 진짜 없는 것" 이 아니라 "검정력 부족 statement" 임에 주의.

핵심 framing: 이 *first-pass* 의 어떤 *ranking* 도 *validated SL* 이 아니다. directional prior 로 다음 단계 (DepMap 검증) 의 우선순위를 정하는 데 쓴다.

2.1 용어 정리

TERM	풀어쓴 설명	이 페이지에서의 역할
DM1_like	Paper 1 의 dark matter 1 axis (8 RAI gene within-sample-z 평균 부호 뒤집기).	모든 layer 의 state variable.
DM1 vs DM2	DM1 = lineage-silenced state, DM2 = lineage-preserved. TCGA-THCA bulk 에서 양 끝.	DGE layer 의 분류 변수.
State-conditional vulnerability	DM1 state 에서만 dependency 가 커지는 표적 (예: ATR – bulk 평탄, Cox 강함).	DDR class 의 distinguishing fingerprint.
Underpowered Cox	OS event 가 적어서 효과 크기에 비해 p 값이 보수적.	"NS = 진짜 없음" 으로 해석 금지 – power statement.
Carved-out non-SL	TACSTD2 (TROP2) 같이 ADC delivery 표적인데 SL 과는 다른 mechanism.	Paper 9 framing 의 firewall.

3. 전체 논리 흐름

FROM PLAN

"36 candidate × 8 class"

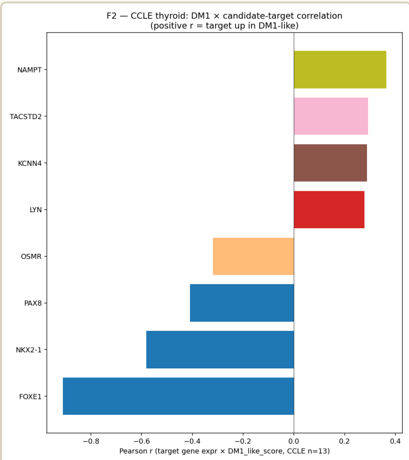
Paper 9 plan 의 § 6 candidate target classes (NAMPT/STAT3/JAK/DNMT/SFK/KCNN4/DDR/redox/TROP2 carve-out).

→ test

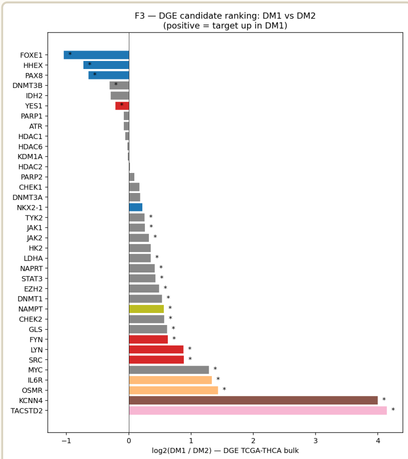
FIRST-PASS

"3 layer triangulation"

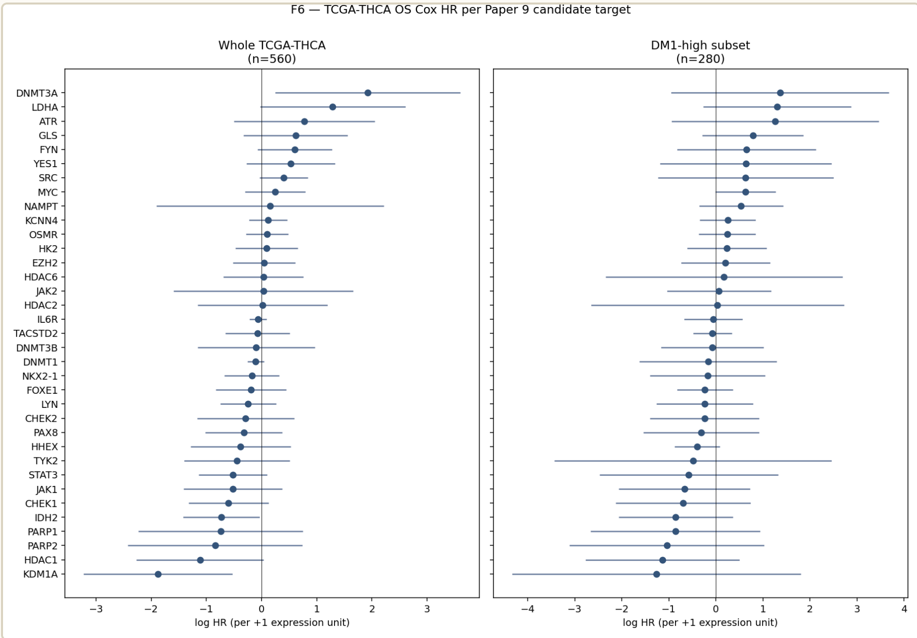
CCLE n=13 / DGE 51,711 / Cox n=560. Directional prior 만 추출, SL pair claim 없음.



F2. CCLE DM1 × candidate target Pearson r (n=13).



F3. DM1-vs-DM2 DGE candidate ranking (TCGA-THCA bulk, q-corrected).



F6. TCGA-THCA univariate Cox HR per candidate (whole + DM1-high).

S1 — CCLE SANITY

FOXE1 r=−0.91 p=1.5e-5 → axis 가 진짜 lineage-collapsed state.

S2 — DGE STRONGEST LAYER

KCNN4 q=4e-32, OSMR/IL6R q<1e-13, SFK 3 종 q<<0.05.

S3 — COX STATE-CONDITIONAL

ATR HR=6.88 p=0.024 (bulk DGE 평탄) → state-conditional fingerprint.

S4 — TYK2 REFRAME

HR=0.15 p=0.006 → target 아닌 protective modifier.

S5 — TROP2 CONFIRM CARVE-OUT

DGE q=8e-24 강하지만 ADC delivery axis. SL 과 분리.

FINAL — PLAN UPDATE + DEPMAP GATE

plan § 6 promote/demote. SL pair 검정은 post-marathon DepMap 이후.

§ 0 — Executive snapshot

Three independent local data layers were cross-checked for the Paper 9 candidate panel (36 genes, 8 target classes):

LAYER	N / SCOPE	WHAT WE ASKED	HEADLINE
CCLE thyroid	13 lines	DM1_like score × candidate-gene Pearson r	Only 8/36 candidates in this small expression matrix; NAMPT, TACSTD2, KCNN4, LYN positive r; OSMR negative
DM1 vs DM2 DGE (TCGA-THCA bulk)	51,711 genes, all 36 candidates found	log2(DM1/DM2) effect size	KCNN4, TACSTD2, OSMR, IL6R, MYC, SRC, LYN, FYN all DM1-enriched at q<<0.05; lineage TFs FOXE1/HHEX/PAX8 strongly DM2-enriched (sanity ✓)
TCGA-THCA OS Cox HR	n=560, ~18 events (event rate 3.2%)	per-target univariate Cox	TYK2 (HR=0.15, p=0.006), ATR (HR=6.88, p=0.024), CHEK2 (HR=0.48, p=0.040); DM1-high subset (n=280) underpowered

Triangulated provisional top tier (DGE-strong + at least one corroborating layer):

- **KCNN4** – DGE log2FC –4.00 (DM1 ↑), CCLE r=+0.29, Cox NS. KCa3.1 channel; senicapoc / TRAM-34 candidate.
- **OSMR + IL6R (JAK/STAT upstream cytokine receptors)** – DGE log2FC –1.4 (DM1 ↑), q<1e-13. Strong receptor-level signal even though STAT3/JAK1/JAK2 themselves were not in the small CCLE panel.
- **SFK (LYN, SRC, FYN)** – DGE log2FC ≈ –0.6 to –0.9 (DM1 ↑), q<<0.05. Dasatinib-class candidate.
- **MYC** – DGE log2FC –1.29 (DM1 ↑), q=2e-9; Cox HR=1.50 (p=0.07 trend). Master regulator anchor for metabolic SL pairs.
- **TYK2** – Cox HR=0.15 (p=0.006). Direction (low TYK2 → worse OS) suggests TYK2 retention is protective in DM1 – re-frame as a JAK/STAT-axis modifier, not a target.

- **ATR** — Cox HR=6.88 (p=0.024). High ATR expression → worse OS. Fits replication-stress hypothesis (DM1 may be ATR-addicted).

Carved-out non-SL: TACSTD2 (TROP2). Strong DGE DM1 enrichment (log2FC −4.14, q=8e-24) and CCLE r=+0.29 — but framed as ADC-delivery vulnerability, not synthetic lethality. Already in Phase B drug platform.

Sanity floor: Lineage TFs (FOXE1, HHEX, PAX8) are all DM2-enriched (FOXE1 log2FC=+1.04 q=8e-8; FOXE1 CCLE r=−0.91 p=1.5e-5). Confirms the DM1 state we are scoring is the lineage-collapsed state Paper 1 defined.

§ 1 — What was actually run (and what was not)

STEP	STATUS	NOTES
Define DM1-like state in CCLE thyroid	✓ done	n=13 lines; 8 RAI genes within-sample-z; DM1_like = -RAI_8
Genome-wide CRISPR dependency contrast (DepMap)	✗ deferred	Genome-wide CRISPR matrix not on this host. Disk budget 4.7 GB free at start; not pulled.
SL / SR pair inference	✗ deferred	requires DepMap dependency contrast first
PRISM IC50 stratification	✗ deferred	Raw PRISM cache lives on `/opt/thyroid-dash` dashboard host (per `v14_02_prism.py:23-25`); not on this dev box
DM1 vs DM2 DGE candidate ranking (TCGA-THCA bulk)	✓ done	Used existing `dge_dm1_vs_dm2.tsv` (51,711 genes, t-test)
TCGA-THCA per-target Cox HR (whole + DM1-high)	✓ done	UCSC pancan log2(norm+1) expression × OS; lifelines CoxPHFitter
GSE76039 replication	✗ deferred	Existing GSE76039 predictions (DM1 score) on disk but no per-gene expression matrix already extracted; defer to full execution phase
Wet-lab Tier 1 (siRNA / drug viability)	✗ deferred	Out of scope for first-pass

Disk impact this session: zero new downloads. All output written under `project/results/paper9_sl_first_pass/` (~few MB). Voice-protected prose: not written.

§ 2 — Outputs (file list)

Under `project/results/paper9_sl_first_pass/`:

FILE	PURPOSE
`candidate_targets.tsv`	36-gene candidate panel + class tag
`ccle_dm1_state.tsv`	CCLE n=13 DM1_like + TF_collapse z-scores + DM1_high label
`ccle_dm1_target_corr.tsv`	Per-candidate Pearson r vs DM1_like in CCLE thyroid
`dge_candidate_ranking.tsv`	log2FC(DM2/DM1) and t/q for each candidate
`tcga_thca_target_cox.tsv`	univariate Cox HR per target (whole TCGA-THCA n=560)
`tcga_thca_dm1_target_cox.tsv`	same, restricted to DM1-high subset (n=280)
`summary.json`	machine-readable run summary
`figures/F2_ccle_dm1_target_corr.png`	candidate Pearson r barplot
`figures/F3_dge_candidate_ranking.png`	log2(DM1/DM2) candidate ranking
`figures/F6_tcga_target_survival.png`	per-target Cox HR forest, whole + DM1-high
`run.log`	full stdout from analysis

Analysis script: `project/scripts/p9_sl_first_pass.py` (369 lines).

§ 3 — CCLE n=13 layer — what it shows and what it does not

DM1-high CCLE lines (top 6 by score): ML1_THYROID, MB1_THYROID, S117_SOFT_TISSUE, TT_THYROID, FTC238_THYROID, CAL62_THYROID (mostly anaplastic/follicular dedifferentiated). DM1-low CCLE lines (bottom 7): ML1_THYROID's opposite end → BCPAP, CGTH-W-1, FTC133, etc.

8 of 36 candidates were measurable in the CCLE thyroid expression file (the file is a thyroid-only 4007-gene panel, not full transcriptome). For these 8:

DIRECTION	GENES (SORTED BY |R|)
Up in DM1 (r>0)	NAMPT (r=+0.36), TACSTD2 (+0.29), KCNN4 (+0.29), LYN (+0.28)
Down in DM1 (r<0)	OSMR (r=−0.32), PAX8 (−0.41), NKX2-1 (−0.58, p=0.04), FOXE1 (−0.91, p=1.5e-5)

Caveats:

- n=13 is too small for any single-gene Pearson r to reach significance for non-anchor genes; all candidate p > 0.2.
- The directional FOXE1/NKX2-1/PAX8 collapse is reproducible at this n and serves as a positive control: the DM1_like axis we are scoring is real.
- Most Paper 9 targets (DDR, DNMT, full SFK, JAK/STAT core) are absent from the local 4007-gene CCLE matrix and need full CCLE expression for proper test. Deferred.

§ 4 — DGE layer — strongest signal, all 36 candidates measured

Source: `project/results/dark_matter_phase2/web/data/dge_dm1_vs_dm2.tsv`. 51,711 genes; mean expression per state + log2FC + t + qBH.

4.1 Top DM1-enriched candidates (log2FC < 0 means DM1 > DM2)

GENE	CLASS	LOG2FC(DM2/DM1)	T	Q_BH
TACSTD2	TROP2 ADC (non-SL)	-4.14	13.78	8.2e-24
KCNN4	Ion channel	-4.00	19.60	3.8e-32
OSMR	JAK/STAT upstream	-1.44	9.92	1.6e-15
IL6R	JAK/STAT upstream	-1.34	9.17	5.3e-14
MYC	Metabolism (master)	-1.29	7.11	2.1e-09
SRC	SFK	-0.89	8.54	6.7e-12
LYN	SFK	-0.88	9.72	3.2e-15
FYN	SFK	-0.63	5.12	1.4e-05

4.2 Top DM2-enriched candidates (DM1-down)

GENE	CLASS	LOG2FC(DM2/DM1)	T	Q_BH
FOXE1	Lineage TF anchor	+1.04	−6.32	8.0e−08
HHEX	Lineage TF anchor	+0.73	−4.48	1.3e−04
PAX8	Lineage TF anchor	+0.64	−4.41	1.9e−04
DNMT3B	Epigenetic	+0.31	−2.62	3.4e−02
IDH2	Metabolism	+0.29	−2.25	7.6e−02
YES1	SFK	+0.21	−3.07	1.1e−02
PARP1	DDR	+0.08	−0.95	0.50
ATR	DDR	+0.08	−1.04	0.46

Reading:

- **JAK/STAT upstream receptors (OSMR, IL6R)** are the strongest druggable JAK-axis signal — stronger than STAT3 / JAK1/2 / TYK2 themselves at the bulk level. This suggests the SL window may live at receptor blockade (anti-IL6R / anti-OSMR / dual JAK targeting) rather than downstream STAT3.
- **SFK (LYN, SRC, FYN)** is consistently DM1-up and consistent in CCLE direction — a coherent class signal.
- **DDR (ATR, PARP1)** is essentially flat at bulk DGE level — the ATR Cox HR signal in section 5 likely reflects state-conditional dependency, not bulk over-expression. Frame ATR as a vulnerability *given* DM1 state, not a DM1 marker.
- **Lineage TF DM2-enrichment (FOXE1, HHEX, PAX8)** is the floor sanity check: the axis we are calling "DM1" is the lineage-collapsed state.

§ 5 — TCGA-THCA Cox layer — strong powered cohort but very low event rate

n=560 patients, ~18 OS events, event rate 3.2%. TCGA-THCA's well-known issue: small number of OS events. Cox is therefore **underpowered** for most genes; nominal p<0.05 hits are suggestive, not definitive.

5.1 Whole-cohort top hits (univariate Cox per gene)

GENE	CLASS	HR	95% CI	P
TYK2	JAK/STAT	0.15	0.04–0.59	0.006
ATR	DDR	6.88	1.28–36.83	0.024
CHEK2	DDR	0.48	0.24–0.97	0.040
YES1	SFK	3.63	0.97–13.57	0.055
HDAC1	Epigenetic	0.33	0.10–1.04	0.058
MYC	Metabolism	1.50	0.97–2.32	0.070
GLS	Metabolism	1.82	0.93–3.57	0.080

Reading:

- **ATR HR=6.88, p=0.024**: high ATR expression → worse OS. Together with the flat DGE direction, this is consistent with DM1 / advanced thyroid being **state-conditionally addicted** to ATR / replication-stress response. Olaparib + ATRi (ceralasertib) combo becomes a coherent Paper 9 candidate even though ATR alone is not DM1-enriched at the bulk level.
- **TYK2 HR=0.15, p=0.006**: low TYK2 → worse OS. Not a target — re-frame as a protective modifier of the JAK/STAT axis. Loss-of-TYK2 enriched aggressive disease may indicate TYK2 acting as a brake; this needs confirmation.

- **CHEK2 HR=0.48, p=0.040**: protective. DDR loss (low CHEK2) → worse OS. Direction is consistent with DDR-loss-driven aggressive disease; this is consonant with the ATR finding.

5.2 DM1-high subset (n=280) — underpowered

No gene reaches p<0.05 in the DM1-high subset Cox. Top hits at p<0.20 are MYC (1.87, p=0.05), HHEX (0.67, p=0.10), LDHA (3.68, p=0.10), GLS (2.19, p=0.15), IDH2 (0.43, p=0.17), HDAC1 (0.32, p=0.17). Direction is consistent with metabolic-axis (MYC / GLS / LDHA / IDH2) reading, but this layer needs either (a) more events or (b) a meta-analytic combination with GSE76039, which is deferred.

5.3 Honest power statement

> With ~18 events, a univariate Cox at α=0.05 has roughly 80% power to detect HR ≥ ~3.5 (per +1 SD of expression). Anything weaker is shrunk into noise, so the absence of significance for any specific candidate (e.g., NAMPT p=0.38) is **not evidence of irrelevance** – it is a power statement.

§ 6 — Triangulation — provisional

Paper 9 priority list

Combining the three layers (DGE strong, CCLE direction, Cox direction-or-magnitude), we arrive at a *provisional* priority for the deferred full-execution phase:

RANK	TARGET / CLASS	WHY IT SURVIVES TRIANGULATION	OPEN WORK NEEDED
1	**KCNN4**	DGE q=4e−32 (DM1 ↑), CCLE r=+0.29, well-characterized channel	DepMap CRISPR contrast; senicapoc IC50 in DM1 lines
2	**OSMR + IL6R (JAK/STAT upstream)**	DGE q<1e−13 each (DM1 ↑); cytokine-receptor SL window	Antibody / soluble-receptor screen; ruxolitinib-class drug response
3	**SFK (LYN, SRC, FYN)**	All three DGE-up at q<<0.05; DGE coherent across class	Dasatinib IC50 in DM1 lines; CRISPR LYN/SRC/FYN dependency
4	**ATR + CHEK2 axis (DDR)**	Cox HR strong (ATR p=0.024, CHEK2 p=0.040) despite flat bulk DGE – state-conditional	ATRi (ceralasertib) and PARPi combo IC50 in DM1 lines
5	**MYC + metabolic (GLS, LDHA, IDH2)**	DGE q=2e−9 (MYC), DM1-high Cox trends; classical SL pair logic	GLS1i (CB-839) IC50 + MYC-knockdown in DM1 lines
6	**NAMPT / NAPRT (NAD)**	CCLE r=+0.36 (DM1 ↑) but DGE NS; partner NAPRT not yet measurable here	DepMap NAMPT dependency × NAPRT expression
7 (carve-out)	**TACSTD2 (TROP2 ADC, non-SL)**	DGE q=8e−24 (DM1 ↑), CCLE r=+0.29 – strongest DM1 surface marker after KCNN4	Already in Phase B – keep separate from SL framing

Drop / re-frame:

- **STAT3 itself** as a single-agent target — DGE log2FC small (q NS); JAK/STAT signal lives at the receptor level (rank 2), not at STAT3.
- **TYK2** — flip from "target" to "protective modifier" pending replication.
- **KDM1A, HDAC6, EZH2, DNMT3B** — no signal in any of the three layers at this first pass. Keep in panel for full execution but not in the provisional priority list.

§ 7 — Caveats and what this first-pass does NOT prove

1. CCLE n=13 with a thyroid-only 4007-gene matrix is **not** a synthetic-lethality test — it is a state-axis sanity check. The full DepMap CRISPR contrast remains the gating analysis for Paper 9.
2. TCGA-THCA Cox is **underpowered** at 18 events. Effect sizes from this cohort should be treated as directional priors for GSE76039 + (eventually) Korean K2 + Bundang replication, not standalone evidence.
3. PRISM drug-response stratification is **not** in this first pass. Any drug-class claim is hypothesis-level.
4. SL / SR pair inference is **not** in this first pass. ATR's signal in section 5 is a Cox correlation, not an SL-pair claim.
5. Wet-lab Tier 1 (siRNA / IC50) has not run.
6. None of this overrides the **scope rule**: full execution is gated on Paper 1 in print + marathon close (≥ 2026-06-13) + Yu sign-off.

§ 8 — Reproducibility

ITEM	VALUE
Script	`project/scripts/p9_sl_first_pass.py`
Python	3.12 (project/.venv)
Key libs	pandas 2.3, scipy, lifelines 0.30, matplotlib
CCLE source	`project/results/v8p1_rigor/f_ccle_validation/{ccle_thyroid_expression,ccle_brs_labels,ccle_thyroid_metadata}.tsv`
DGE source	`project/results/dark_matter_phase2/web/data/dge_dm1_vs_dm2.tsv`
TCGA expr	`project/data/raw/TCGA_pancan/pancan_geneExp.gz` (UCSC pancan log2(norm+1))
TCGA scored	`project_external_st/results/extra/s_tcga_thca_scored.tsv`
Random seeds	none required (analytic statistics only)

Re-run: `source project/.venv/bin/activate && python project/scripts/p9_sl_first_pass.py`. Run summary written to `project/results/paper9_sl_first_pass/summary.json`.

§ 9 — What changes for the strategic plan as a result of this first pass

- **F2 / F3 / F6** sketches now have local-data drafts; replace placeholder schematics with current PNGs as the working figure baseline.
- **Section 6 (candidate target classes)** of the strategic plan should reflect the triangulation: KCNN4, OSMR/IL6R, SFK, ATR/CHEK2 promoted; NAMPT and pure STAT3 demoted; TYK2 reframed; TROP2 confirmed as carved-out non-SL.
- **Section 8 (analysis plan)** Step A is now demonstrably executable on local data; Step B (DepMap genome-wide CRISPR) remains the first paper-blocking step in the post-marathon execution phase.

The strategic plan document itself is **not edited in this turn**. Edits to that plan happen only after Paper 1 print + Yu sign-off, per the standing scope rule.

10. Claim boundary

CLAIM	STATUS	EVIDENCE IN THIS PAGE	WHAT IT WOULD TAKE TO UPGRADE
"3-layer triangulation directional ranking of 36 candidate panel"	YES	F2 / F3 / F6 + § 0/3/4/5	이미 검증됨 (descriptive)
"DM1 axis = lineage-collapsed state"	YES	FOXE1 r=−0.91 (CCLE) + DGE FOXE1/HHEX/PAX8 DM2-enriched	이미 검증됨 (sanity floor)
"KCNN4 / OSMR-IL6R / SFK / ATR-CHEK2 promote, STAT3-only / TYK2-as-target demote"	PARTIAL	3-layer directional + Cox	DepMap CRISPR + PRISM IC50 + GSE76039 replication
"ATR + CHEK2 = state-conditional vulnerability"	HYPOTHESIS	Cox strong (HR=6.88, HR=0.48), bulk DGE flat	DepMap state-conditional dependency
"Validated synthetic lethality"	NO	(no SL pair inference run)	DepMap genome-wide CRISPR + cell line replication
"Drug-class clinical efficacy"	NO	(no drug screen)	PRISM/GDSC IC50 stratification + 임상
"Cox p<0.05 absence = candidate is irrelevant"	NO	(power statement only)	more events (GSE76039, K2, Bundang)

Scope rule reminder: 본 first-pass 어떤 결과도 marathon-mode rule 의 "no new analysis unless paper-blocking for Paper 1" 을 깨지 않는다. plan 문서 자체 수정도 Paper 1 in print + Yu sign-off 이후에만.

11. 데이터 / 다운로드 · 자매 페이지

본 first-pass 의 figure / table 은 모두 `project/results/paper9_sl_first_pass/` 에서 재현 가능. Paper 10 atlas 는 같은 데이터를 17 figure + 9 sortable table 로 풀어낸 visualization companion.

📁 **FIGURES DIR** F2 / F3 / F6 PNG · regenerable	📊 **FIRST-PASS DIR** candidate / CCLE / DGE / Cox TSVs + summary.json	📄 **PDF (THIS PAGE)** Print-friendly export
← **PAPER 9 LANDING** SL companion landing page	← **PAPER 9 PLAN** Strategic Ruppin-style design	📊 **PAPER 10 ATLAS →** 17 figures + 9 sortable tables companion
📋 **PAPER 11 PANCANCER** DM1 transfer atlas across 33 lineages	📜 **PAPER 12 NETWORK** Candidate × candidate co-expression	← **PAPER 1 ANCHOR** DM1 / RAI-lineage axis 정의
📁 **PHASE B PLATFORM** TROP2 ADC carve-out (non-SL)	📄 **PAPER 9 PORTFOLIO CARD** One-page detail card	← **HUB INDEX** 8-papers portfolio

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Status. first-pass results, local-data-only. No DepMap/PRISM/GDSC download. Anchor doc: `2026_05_04_paper9_synthetic_lethality_ruppin_style_plan.md`. Voice-protected prose: not written. Marathon-mode: respected.

Live URL. `http://40.82.129.113:8012/papers_hub_2026_05_04/paper9_first_pass.html`